

PHIL 350: Final

Andrew Bacon

A three valued semantics for the language $\mathcal{L}(\neg, \rightarrow)$ is given as follows. Is 3-valued truth value assignment is a function v_0 mapping letters to either 0, $\frac{1}{2}$ or 1. A 3-valued valuation is defined by recursion as follows

$$v(P) = v_0(P)$$

$$v(\neg A) = 1 - v(A)$$

$$v(A \rightarrow B) = \min(1, 1 - v(A) + v(B))$$

Truth-tables for the connectives are given on the last page of the exam.

1. Suppose that v_0 is a 3-valued truth-value assignment that assigns every letter either 1 or 0, and let v be the corresponding 3-valued valuation. (recall that the standard definition of an ordinary valuation looks exactly like our definition above, except the last line is: $v'(A \rightarrow B) = \max(1 - v'(A), v'(B))$)
 - (a) Note that since v_0 only takes 1 or 0 as values, it is an ordinary (2-valued) truth-value assignment, so let v' be the corresponding ordinary (i.e. 2-valued) valuation. Write out the recursive definition of v' as we gave it in class, and note any differences from the above definition of a 3-valued valuation. (5 points).
 - (b) Prove by induction that $v(A) = v'(A)$ for all sentences A in $\mathcal{L}(\neg, \rightarrow)$. (15 points)
 - (c) A set of connectives is truth-functionally incomplete *with respect to 3-valued semantics* if there is a 3-valued connective that cannot be defined. Prove that there is a unary 3-valued truth function that cannot be defined. A unary truth function looks like the table for $\neg$ at the end of the exam, except the righthand column can be defined in any way you like. (10 points).

Let us say that $A_1, \dots, A_n \models_3 B$, pronounced “ B is a 3-valued consequence of $A_1, \dots, A_n$ ” when, for every 3-valued valuation v , if $v(A_1) = \dots = v(A_n) = 1$, then $v(B) = 1$. A is a 3-valued tautology iff $v(A) = 1$ for every 3-valued valuation v .

An axiom system S is sound for 3-valued semantics when the following holds: If $A_1, \dots, A_n \vdash_S B$ then $A_1, \dots, A_n \models_3 B$. In the next question you will show that the following axiom system, S , is sound for 3-valued semantics.

A1. $A \rightarrow (B \rightarrow A)$

A4. $(A \rightarrow B) \rightarrow ((B \rightarrow C) \rightarrow (A \rightarrow C))$

A3. $(\neg A \rightarrow \neg B) \rightarrow (B \rightarrow A)$

Modus Ponens From A and $A \rightarrow B$ infer B

2. (a) Show that A1 and A3 are 3-valued tautologies. (10 points)
- (b) Suppose that $C_1 \dots, C_n \models_3 A$, and $C_1 \dots, C_n \models_3 (A \rightarrow B)$. Show that it follows that $C_1 \dots, C_n \models_3 B$. (10 points).
- (c) Prove by induction on proofs that $C_1, \dots, C_n \models_3 A$ for every line A of a proof in S from assumptions $C_1, \dots, C_n$. Explain how this gives 3-valued soundness of S . You may use previous questions and you may assume that A4 is a 3-valued tautology. (20 points).

In this final question you will show that the axiom system S described above is incomplete in the ordinary sense (i.e. incomplete with respect to the ordinary 2-valued notion of entailment.)

3. (a) Find a sentence of the language $\mathcal{L}(\neg, \rightarrow)$ that is a 2-valued tautology that is not a 3-valued tautology. (15 points).
- (b) State what it means for S to be complete (in the ordinary sense, with respect to regular 2-valued entailment). Explain why part a implies that S is incomplete. (15 points).

P	Q	$P \rightarrow Q$	P	$\neg P$
1	1	1		
1	$\frac{1}{2}$	$\frac{1}{2}$	1	0
1	0	0	$\frac{1}{2}$	$\frac{1}{2}$
$\frac{1}{2}$	1	1	$\frac{1}{2}$	1
$\frac{1}{2}$	$\frac{1}{2}$	1	0	1
$\frac{1}{2}$	0	$\frac{1}{2}$		
0	1	1		
0	$\frac{1}{2}$	1		
0	0	1		